

# Lab 08 Template

- 1.) Screenshot of the default Apache page via Lynx  
**www.studentXX.230.com**  
(10 points)

```

                                Apache2 Ubuntu Default Page: It works (p1 of 3)

Ubuntu Logo
Apache2 Default Page
It works!

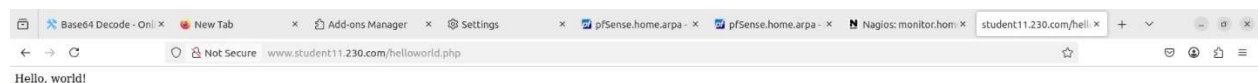
This is the default welcome page used to test the correct operation of the Apache2 server
after installation on Ubuntu systems. It is based on the equivalent page on Debian, from
which the Ubuntu Apache packaging is derived. If you can read this page, it means that the
Apache HTTP server installed at this site is working properly. You should replace this file
(located at /var/www/html/index.html) before continuing to operate your HTTP server.

If you are a normal user of this web site and don't know what this page is about, this
probably means that the site is currently unavailable due to maintenance. If the problem
persists, please contact the site's administrator.
Configuration Overview

Ubuntu's Apache2 default configuration is different from the upstream default
configuration, and split into several files optimized for interaction with Ubuntu tools.
The configuration system is fully documented in /usr/share/doc/apache2/README.Debian.gz.
Refer to this for the full documentation. Documentation for the web server itself can be
found by accessing the manual if the apache2-doc package was installed on this server.

The configuration layout for an Apache2 web server installation on Ubuntu systems is as
follows:
/etc/apache2/
|-- apache2.conf
|   |-- ports.conf
|-- mods-enabled
|   |-- *.load
|   |-- *.conf
|-- conf-enabled
|   |-- *.conf
|-- sites-enabled
|   |-- *.conf
-- press space for next page --
Arrow keys: Up and Down to move. Right to follow a link; Left to go back.
H)elp O)ptions P)rint G)o M)ain screen Q)uit /=search [delete]=history list
```

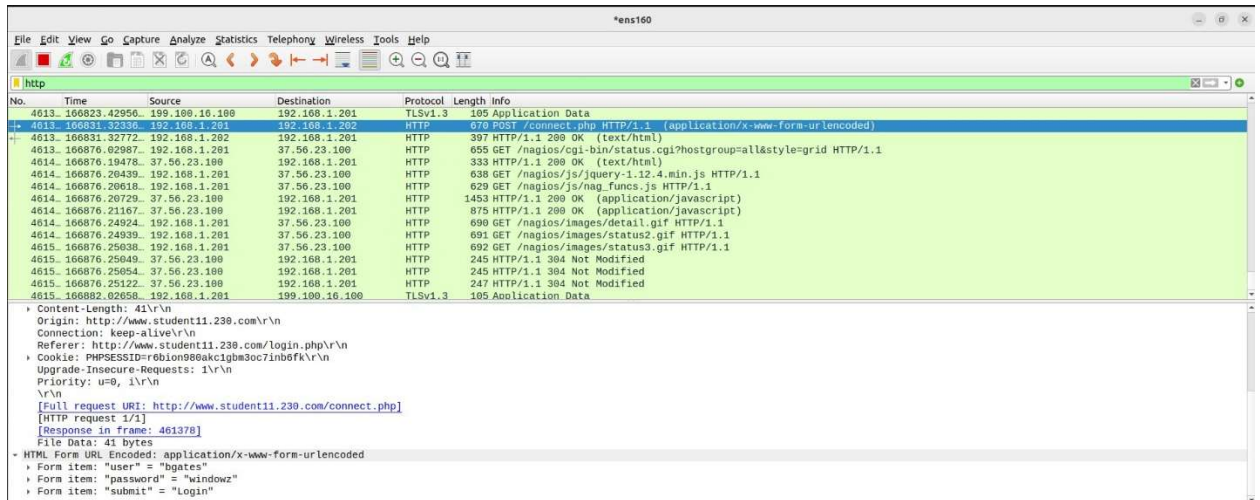
- 2.) Screenshot of the Hello, world! PHP page via Firefox  
**www.studentXX.230.com/helloworld.php**  
(10 points)



- 3.) Screenshot of one of the users logged into PHP web page via Firefox  
**www.studentXX.230.com/login.php**  
(10 points)



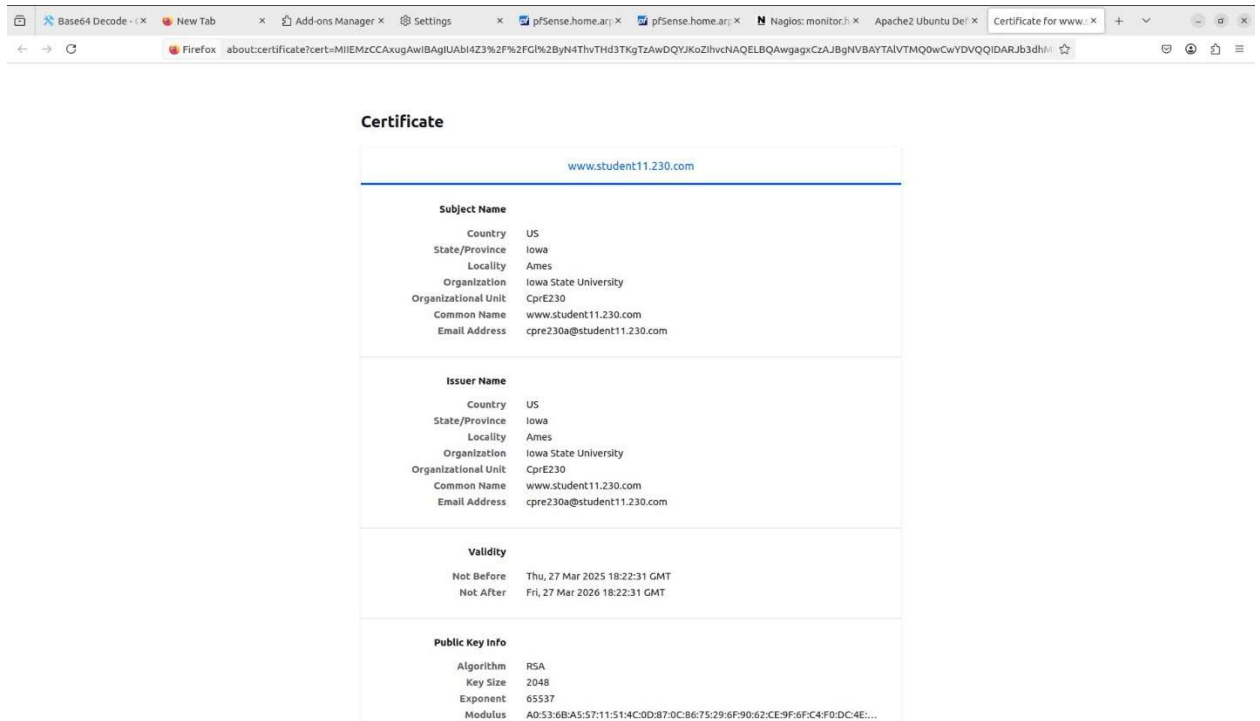
- 4.) Screenshot of Wireshark showing the plaintext password when logging in  
(5 points)



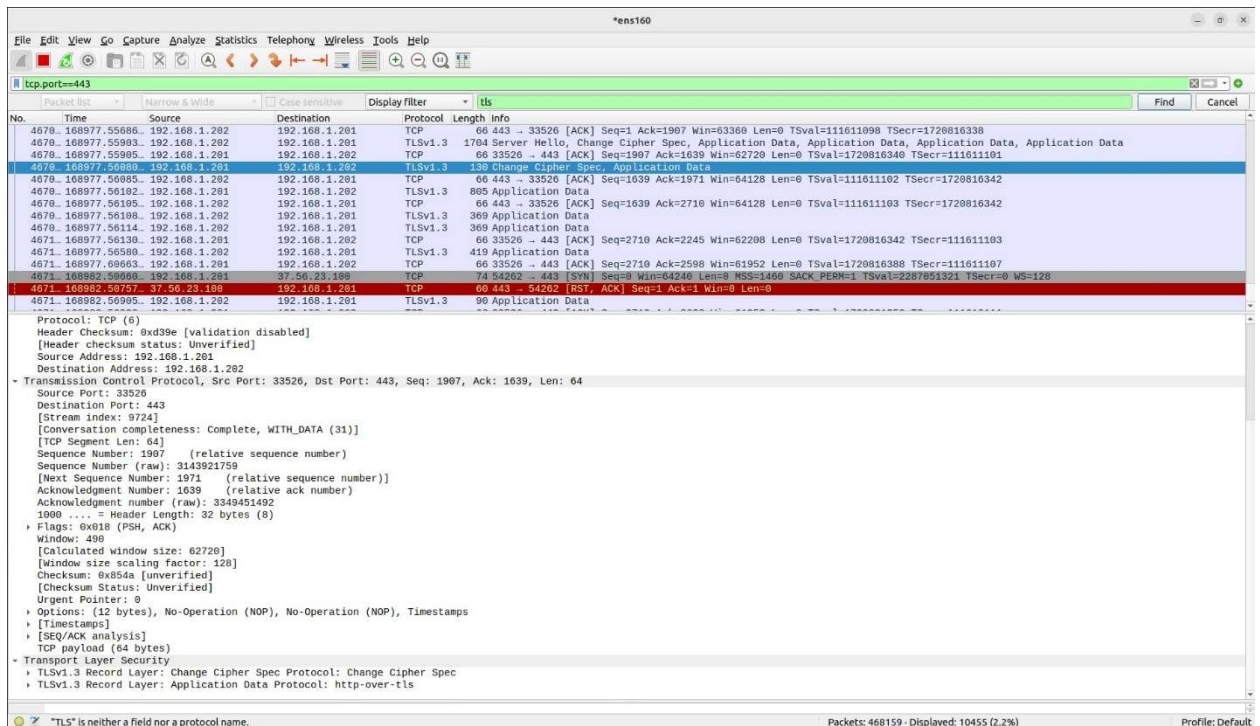
5.) Screenshot of the Apache default page on HTTPS via Firefox  
<https://www.studentXX.230.com>  
(10 points)



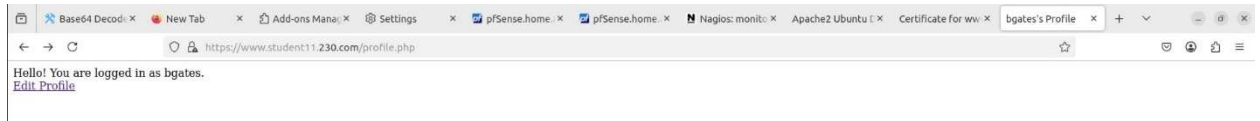
## 6.) Screenshot of the SSL certificate (10 points)



## 7.) Screenshot of Wireshark showing an encrypted packet when logging in (5 points)



- 8.) **Screenshot of the profile page, via Firefox**  
**<https://studentXX.230.com/profile.php>**  
(10 points)



- 9.) **Screenshot of the database, with changed username and password**  
(5 points)

```
cpre230@www:~$ sudo mysql
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 37
Server version: 8.0.41-0ubuntu0.22.04.1 (Ubuntu)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> select * from login_info.UsernamePassword;
+-----+-----+-----+
| usernameID | username | password |
+-----+-----+-----+
|          1 | testxyz  | apple=!(doctor) |
|          2 | bgates   | windowz |
|          3 | mzuckerberg | IAmNotARobot |
|          4 | emusk    | i<3XAEAX-ii |
|          5 | jbezos   | Money^2 |
+-----+-----+-----+
5 rows in set (0.00 sec)

mysql>
```

10) **Description of what SQL injection is, and how you might go about preventing it**  
(10 points)

Ans) Sql injection is basically a web security vulnerability that allows an attacker to modify or delete queries that an application makes to its database. These queries might include data that belongs to other users as well. We can prevent most sql injections by using parameterized queries. These queries can be used in any situation where untrusted input appears as data within query, including the WHERE clause and the in an INSERT or UPDATE statement. In order to make a parameterized query effective, the string that is used in the query should always be a hard coded constant, it must never contain any variable data from any origin. (Source: <https://portswigger.net/web-security/sql-injection>)

11) **Screenshot of NAT rule(s)**  
(5 points)

Port Forward

1:1

Outbound

NPT

Rules

| <input type="checkbox"/> | Interface                               | Protocol | Source Address | Source Ports | Dest. Address | Dest. Ports      | NAT IP        | NAT Ports        | Description                             | Actions |
|--------------------------|-----------------------------------------|----------|----------------|--------------|---------------|------------------|---------------|------------------|-----------------------------------------|---------|
| Desktop - Nagios check   |                                         |          |                |              |               |                  |               |                  |                                         |         |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> WAN | TCP      | *              | *            | 42.49.30.201  | 22 (SSH)         | 192.168.1.201 | 22 (SSH)         | desktop - Nagios ssh check              |         |
| NS2 - Nagios checks      |                                         |          |                |              |               |                  |               |                  |                                         |         |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> WAN | TCP/UDP  | 37.56.23.100   | *            | 42.49.30.199  | 53 (DNS)         | 192.168.1.200 | 53 (DNS)         | ns2 - Nagios DNS for inside machines    |         |
| Idap-Nagios checks       |                                         |          |                |              |               |                  |               |                  |                                         |         |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> WAN | TCP      | 37.56.23.100   | *            | 42.49.30.205  | 389 (LDAP)       | 192.168.1.205 | 389 (LDAP)       | Idap - Nagios checks                    |         |
| Mail-Nagios checks       |                                         |          |                |              |               |                  |               |                  |                                         |         |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> WAN | TCP      | 37.56.23.100   | *            | 42.49.30.204  | 993 (IMAP/S)     | 192.168.1.204 | 993 (IMAP/S)     | Allow IMAP access                       |         |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> WAN | TCP      | *              | *            | 42.49.30.204  | 587 (SUBMISSION) | 192.168.1.204 | 587 (SUBMISSION) | Allow SMTP submission                   |         |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> WAN | TCP      | *              | *            | 42.49.30.204  | 110 (POP3)       | 192.168.1.204 | 110 (POP3)       | Mail pop                                |         |
| www - Nagios checks      |                                         |          |                |              |               |                  |               |                  |                                         |         |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> WAN | TCP      | *              | *            | 42.49.30.202  | 80 (HTTP)        | 192.168.1.202 | 80 (HTTP)        | Allow external web access to web server |         |
| <input type="checkbox"/> | <input checked="" type="checkbox"/> WAN | TCP      | *              | *            | 42.49.30.202  | 443 (HTTPS)      | 192.168.1.202 | 443 (HTTPS)      | Allow external web access to web server |         |

Add

Add

Delete

Save

Separation



12) **Screenshot of successful Nagios queries against web services.**  
(10 points)

| Student 11 (student11)     |                                                                                                     |                                                                                                                                                                                                                                                             |
|----------------------------|-----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Host                       | Services                                                                                            | Actions                                                                                                                                                                                                                                                     |
| desktop1.student11.230.com | desktop1 ssh                                                                                        |    |
| ldap.student11.230.com     | ldap Bezos ldap Gates ldap Jobs ldap Musk ldap Zuckerberg ldap service                              |    |
| mail.student11.230.com     | dns MX mail pop secure imap secure imap login secure smtp                                           |    |
| ns1.student11.230.com      | dns mail dns ns1 dns www dns www2 reverse dns mail reverse dns ns1 reverse dns www reverse dns www2 |    |
| ns2.student11.230.com      | dns desktop1 dns ldap dns ws reverse dns desktop1 reverse dns ldap reverse dns ws                   |    |
| www.student11.230.com      | www http www https www login                                                                        |    |
| www2.student11.230.com     | www2 django www2 ssh                                                                                |    |